

Organochlorine Exposure and Incidence of Diabetes in a Cohort of Great Lakes Sport Fish Consumers

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Background Studies have demonstrated ubiquitous human exposure to persistent organic pollutants (POPs) such as p,p'-diphenyldichloroethene (DDE) and polychlorinated biphenyls (PCBs). Although there is considerable evidence that POP exposures are associated with prevalent diabetes, these studies do not establish causality because the cross-sectional study design does not allow for assessment of temporality of the exposure-disease association. Prospective studies, however, have been lacking. Objectives This study was designed to determine whether POP body burdens are related to incidence of diabetes in a cohort of Great Lakes sport fish consumers. Methods The cohort was established in the early 1990s and followed through 2005. We tested serum for DDE and PCB congeners and assessed diabetes diagnosis, demographics, and fish consumption. Associations of diabetes with exposures were examined prospectively in participants without diabetes in 1994–1995, followed through 2005. Annual percent changes in DDE and PCB-132/153 from 1994 to 2005 were examined by diabetes status. Results DDE exposure was associated with incident diabetes. Incident diabetes was not associated with mono-ortho PCB-118, total PCBs, or years of sport fish consumption. Annual percent change in DDE and PCB-132/153 did not differ significantly by diabetes status. Conclusions This study demonstrates an association between DDE exposure and incident diabetes. The findings of an association of DDE with incident diabetes and the lack of effect of diabetes on annual percent change in POPs do not support the hypothesis that associations of POPs with diabetes are attributable to reverse causality. Additional studies should address the biological pathways by which DDE could affect glucose homeostasis.